

## Australia's Bridge Asset Intelligence Platform

End-to-end lifecycle management for bridge assets across Australian states — from GIS inspection to NHVR heavy-vehicle compliance.

### Asset Registry

Full CRUD on 5,000+ bridge records across 8 states. Rich attributes: location, structure, condition, load ratings.

### GIS Map View

Interactive Leaflet map with condition heat-map, restriction overlays, and viewport-aware API fetching.

### Inspections & Conditions

Field condition capture with photo attachments. Condition trends tracked over time per bridge.

### NHVR Compliance

Heavy-vehicle route assessment, gazette expiry tracking, and approval workflow for Class 1–3 vehicles.

### Mass Upload

Excel / CSV bulk import with validation preview. Supports all datasets via MTA endpoint.

### Analytics Dashboard

Network KPIs, condition distribution, restriction summary, and trend charts updated in real time.

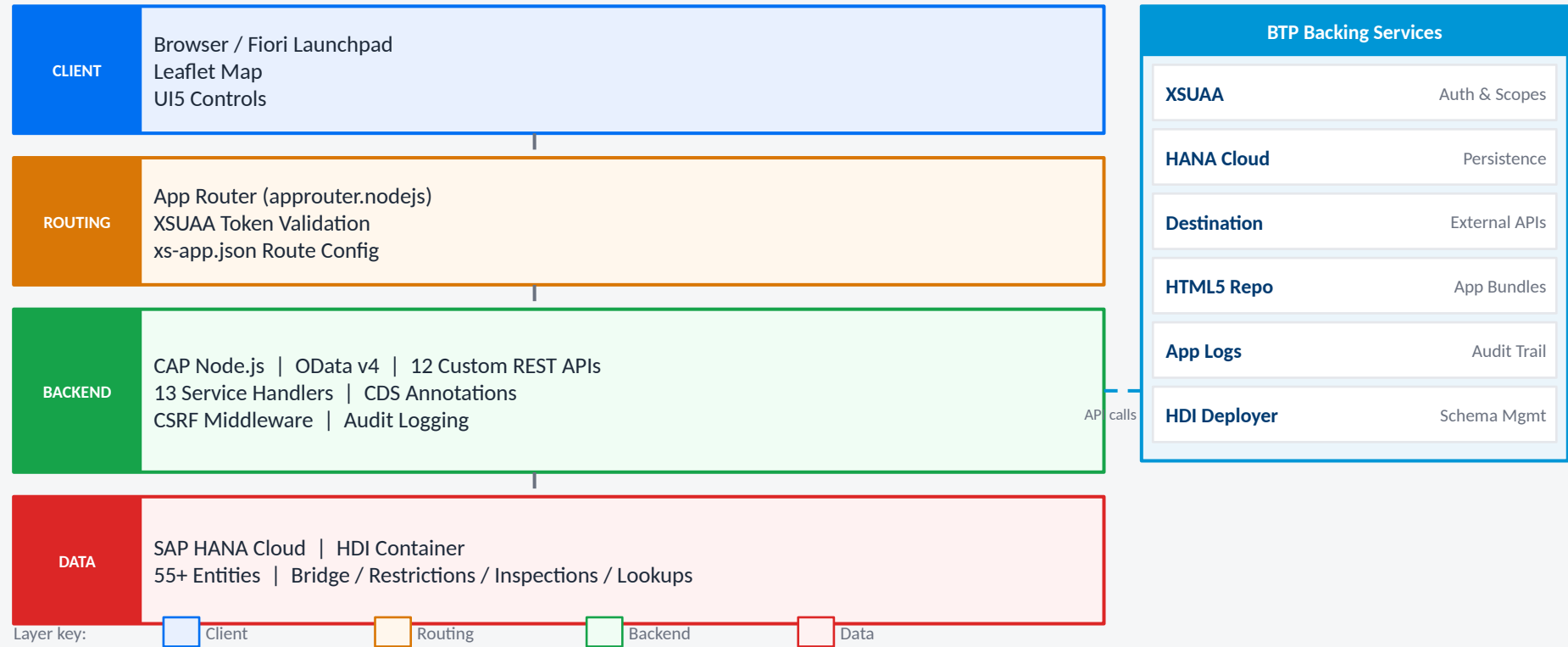
**5,000+** Bridge Records

**8** Australian States

**13** CAP Services

**10** Fiori Apps

**5** User Roles



| BTP Service            | Plan                   | What It Enables in BMS                                                                                             |
|------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------|
| SAP HANA Cloud         | hdi-shared             | Persists all bridge, restriction, inspection, and lookup data. HDI deployer manages schema migrations.             |
| XSUAA                  | application            | Enforces role-based access (5 scopes, 6 role-templates). Issues JWT tokens consumed by CAP and App Router.         |
| HTML5 App Repository   | app-host + app-runtime | Hosts and serves 10 Fiori app bundles (admin, map, mass-upload, operations, dashboards) without custom web server. |
| Application Router     | approuter.nodejs       | Single entry point for all apps. Routes traffic, validates XSUAA tokens, and injects X-CSRF-Token.                 |
| Destination Service    | lite                   | Manages named connections to NHVR external API, state transport APIs, and WMS tile endpoints.                      |
| Application Logging    | lite                   | Captures all CAP audit logs, change-log events, and custom Express router access logs for BTP log viewer.          |
| Alert Notification Svc | standard (optional)    | Triggers email/webhook alerts when bridge condition rating falls below configured threshold.                       |
| Cloud Foundry Runtime  | —                      | Hosts BridgeManagement-srv (CAP Node.js) and BridgeManagement approuter as CF apps.                                |

| Role Collection    | XSUAA Scope | Capabilities                                                                      |
|--------------------|-------------|-----------------------------------------------------------------------------------|
| BMS Administrator  | .admin      | Full CRUD: bridges, restrictions, lookups, mass upload, system config, audit logs |
| BMS Bridge Manager | .manage     | Create & edit bridges, restrictions, attachments, condition capture, reports      |
| BMS Operator       | .operate    | Read + mass upload, mass edit, NHVR compliance, GIS map write                     |
| BMS Inspector      | .inspect    | Read + condition capture, inspection creation, defect logging                     |
| BMS Viewer         | .view       | Read-only: map, reports, dashboards. No write access anywhere.                    |





## Routine Tasks

- Seed data refresh via cds deploy after lookup table edits
- Schema migrations: edit db/schema → cds deploy --to hana
- NHVR gazette re-assessment when state regulations change
- Mass upload template regeneration after schema changes



## Backup & Recovery

- HANA Cloud: automated daily backups, 14-day retention
- Point-in-time recovery via BTP cockpit
- HDI container snapshots before major schema migrations



## Operational Notes

- Upload limit: ~18 MB raw file (25 MB JSON limit via express.json). Label in UI says 50 MB — update if raising the limit to 70 MB.
- CSV + 'All Datasets' is rejected server-side — add client-side validation in MassUpload.controller.js.
- Port 8008 local (4004 reserved by adjacent project). BTP uses standard CF ports.
- Run `npm run validate-build` (`cds build` + `mbt build`) before every BTP deployment.



## Monitoring

- BTP Application Logging: CAP audit logs + REST access logs
- Alert Notification: condition rating degradation triggers
- Health endpoint: GET /health — returns {status: 'ok'}
- @cap-js/audit-logging captures all entity mutations



## Version Cadence

- @sap/cds v9+ — track monthly SAP CAP release notes
- Node.js 20 LTS required (enforce via .nvmrc: 20)
- xlsx 0.18.x — stable; avoid major version upgrades



## MTA Modules

| Module                       | Type             | Notes                                       |
|------------------------------|------------------|---------------------------------------------|
| BridgeManagement-srv         | nodejs           | CAP backend — OData v4 + 12 REST APIs       |
| BridgeManagement (router)    | approuter.nodejs | Token validation, routing, CSRF             |
| BridgeManagement-db-deployer | hdb              | HDI schema deployment to HANA               |
| 10 × HTML5 apps              | html5            | admin, map, mass-upload, ops, dashboards... |

## Key Commands

| Command                                       | Purpose                         |
|-----------------------------------------------|---------------------------------|
| <code>npm start</code>                        | Local dev server (port 8008)    |
| <code>npm test</code>                         | Jest integration tests (SQLite) |
| <code>cds deploy --to sqlite:db.sqlite</code> | Validate schema locally         |
| <code>cds deploy --to hana</code>             | Migrate HANA Cloud schema       |
| <code>npm run validate-build</code>           | cds build + mbt build check     |
| <code>mbt build --mtar BMS.mtar</code>        | Produce deploy artifact         |
| <code>cf deploy BMS.mtar</code>               | Push all modules to BTP CF      |

### CF Spaces:

dev — feature branches, local parity

staging — UAT + regression testing

prod — live — protected, manual gate